Box 51.1 Metabolic Effects, Symptoms, and Signs of Uremia

Metabolic

Increased oxidant levels
Reduced resting energy expenditure
Reduced body temperature
Insulin resistance
Muscle wasting
Amenorrhea and sexual dysfunction

Neural and Muscular

Fatigue
Altered mental status ranging from loss of concentration to coma and seizures
Sleep disturbances
Restless legs
Peripheral neuropathy
Anorexia and nausea
Diminution in taste and smell
Itching
Cramps
Reduced muscle membrane potential

Other

Serositis (including pericarditis)
Hiccups
Granulocyte and lymphocyte dysfunction
Platelet dysfunction
Shortened erythrocyte life span
Albumin modifications

Box_51_1

Box 51.2 Uremic Abnormalities Transferable With Uremic Serum or Plasma

Inhibition of Na^+,K^+-ATPase
Inhibition of platelet function
Leukocyte dysfunction
Loss of erythrocyte membrane lipid asymmetry
Insulin resistance

ATPase, Adenosine triphosphatase.

Box_51_2

Box 51.3 Low-Molecular-Weight Proteins and Protein Fragments That Accumulate in Uremia

α_1-Microglobulin
β_2-Microglobulin
β-Trace protein (also known as prostaglandin D2 synthase)
Clara cell protein
Chromogranin A
Cystatin C
Free immunoglobulin light chains
Natriuretic peptides
Procalcitonin
Retinol-binding protein
Transcobalamin
Leptin
Ghrelin
Resistin
Troponin I

Box_51_3

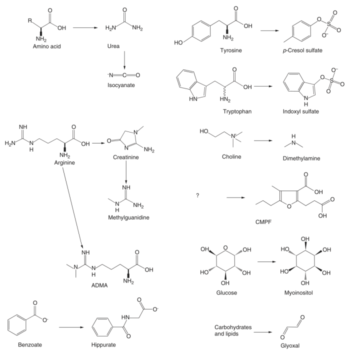

Fig. 51.1 Generation of potential uremic toxins. The substances in the *right column* of each panel are metabolites that are normally excreted by the kidney and therefore accumulate in the extracellular fluid when kidney function is lost. The *left column* shows the substances from which these potential "uremic toxins" are derived. In some cases, the biochemical derivation of the potential toxins is uncertain. For example, it is not known what fraction of the dimethylamine normally excreted is derived from choline, and the source of 3-carboxy-4-methyl-5-prophy-2-furanpropanoic acid (CMPF) is obscure. See text for details. *ADMA,* Asymmetric dimethyl arginine.

Fig_51_1

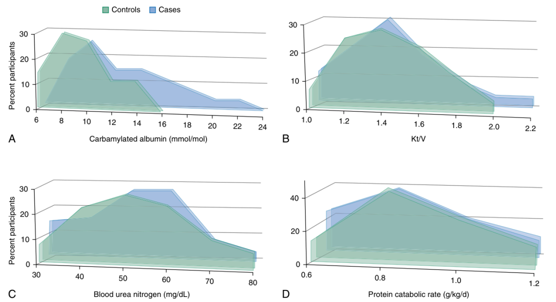

Fig. 51.2 Patient and control distributions of carbamylated albumin, Kt/V, blood urea nitrogen (BUN), and normalized protein catabolic rate. (A) Distribution of carbamylated albumin level measures in the study population by patients (blue; n = 122 who died within 1 year of starting dialysis) and matched controls (green; n = 244 who survived the first year of dialysis); P = 0.01). (B) Distribution of the average Kt/V measure in the study population by patients (blue) and controls (green; P = 0.55). (C) Distribution of the average BUN measure in the study population by patients (blue) and controls (green; P = 0.51). (D) Distribution of normalized protein catabolic rate measured in the study population by patients (blue) and controls (green; P = 0.15). P values represent the frequency distribution comparisons. (Reproduced from Kalim S, Trottier CA, Wenger JB, et al. Longitudinal changes in protein carbamylation and mortality risk after initiation of hemodialysis. *Clin J Am Soc Nephrol.* 2016;11:1809–1816.)

Fig_51_2

Table 51.1 Clinical Studies Linking Urea and Protein Carbamylation to Clinical Outcomes

Study Design	Sample Population	Outcomes	Ref.
Case control	550 with preserved eGFR	MACE	22
Cohort	187 on HD	Mortality	41
Case control	366 on HD	Mortality	42
Cohort	3111 with CKD	CKD progression, mortality	43
Cohort	8040	CKD progression	189
Cohort	347 on HD	Mortality	380
Cohort	111 with CHF	Mortality Cardiac transplantation	381
Cohort	158 on HD	Erythropoietin resistance; mortality	382
RCT post-hoc analysis	1161 on HD	CV death	383
Case control	300 with CKD	CKD progression, mortality	384
Cohort	1320 with CKD	CKD progression	385

CHF, Congestive heart failure; *HD,* hemodialysis; *MACE,* major

Table_51_1